

EV ADOPTION REGISTERED THROUGH WASHINGTON STATE

Team Vizionaries

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Our Mission and Vision

Vision

Our vision is to support the shift toward a cleaner and smarter transportation future by making electric vehicle (EV) data more transparent, understandable, and actionable. We believe that through effective data visualization, we can help bridge the gap between environmental goals and real-world implementation empowering communities, policymakers, and planners to make informed, equitable decisions around EV adoption and infrastructure.

Mission

Our mission is to convert Washington State's extensive EV registration dataset into meaningful visual stories that uncover trends, reveal adoption gaps, and support policy-driven insights. Using advanced tools like time-series charts, sunburst and sankey diagrams, choropleth maps, violin plots, and interactive platforms like Kepler.gl, we aim to explore how vehicle type, cost, range, geography, and policy eligibility shape the EV landscape. Through this, we hope to inspire data-driven planning, encourage sustainable choices, and make complex data accessible to everyone.

Why Washington Dataset?

Washington State stands out as a national leader in electric vehicle (EV) adoption, driven by progressive climate policies, eco-conscious consumers, and expanding charging infrastructure. The state plans to reduce vehicle emissions by 68% by 2030 and will ban the sale of most gas-powered vehicles starting with the 2030 model year. Its unique geographic and demographic diversity from urban hubs like Seattle to rural areas makes it an ideal case for studying how EV adoption varies by region. With over 230,000 EV registrations, Washington's dataset offers rich insights into real-world trends, policy impact, and infrastructure needs making it the perfect foundation for our data visualization project.

Tracks EV Adoption Over Time

Bar and area charts revealed a sharp increase in Battery Electric Vehicle (BEV) adoption, especially after 2018, with peaks in 2022–2023. This temporal trend helps forecast demand and infrastructure needs.

1

Analyzes Affordability & Range

Your histograms, box plots, and violin plots demonstrate how BEVs tend to cost more but also offer significantly higher range than PHEVs. This helps buyers and policymakers balance cost, performance, and incentives.

4

Reveals Regional Disparities

Maps and heatmaps show that EV adoption is concentrated in urban counties like King and Snohomish, while rural areas lag behind. This highlights the need for targeted policies to promote equitable infrastructure development.

2

Why This Project Matters

Exposes Gaps in Clean Fuel Incentives

Choropleth maps and stacked bars around CAFV eligibility reveal how eligibility for clean vehicle incentives varies by region and model—critical for refining public rebate programs.

5

Captures Brand & Model Preferences

Through stacked bar, sunburst, and Sankey diagrams, you identified dominant players (e.g., Tesla) and showed how consumer choices differ by make, model, and EV type. This insight supports manufacturers and market strategists.

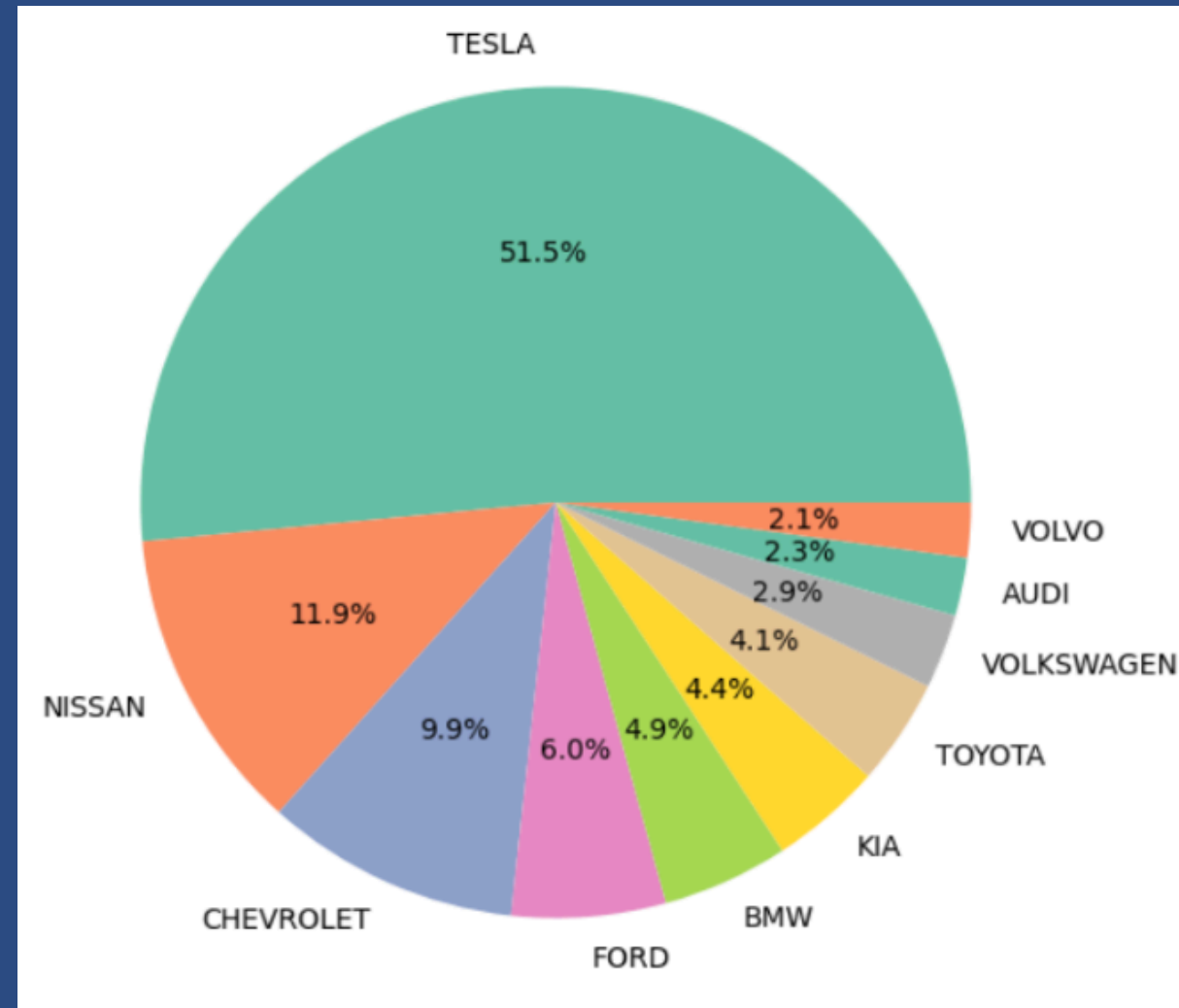
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Improves Visualization Standards

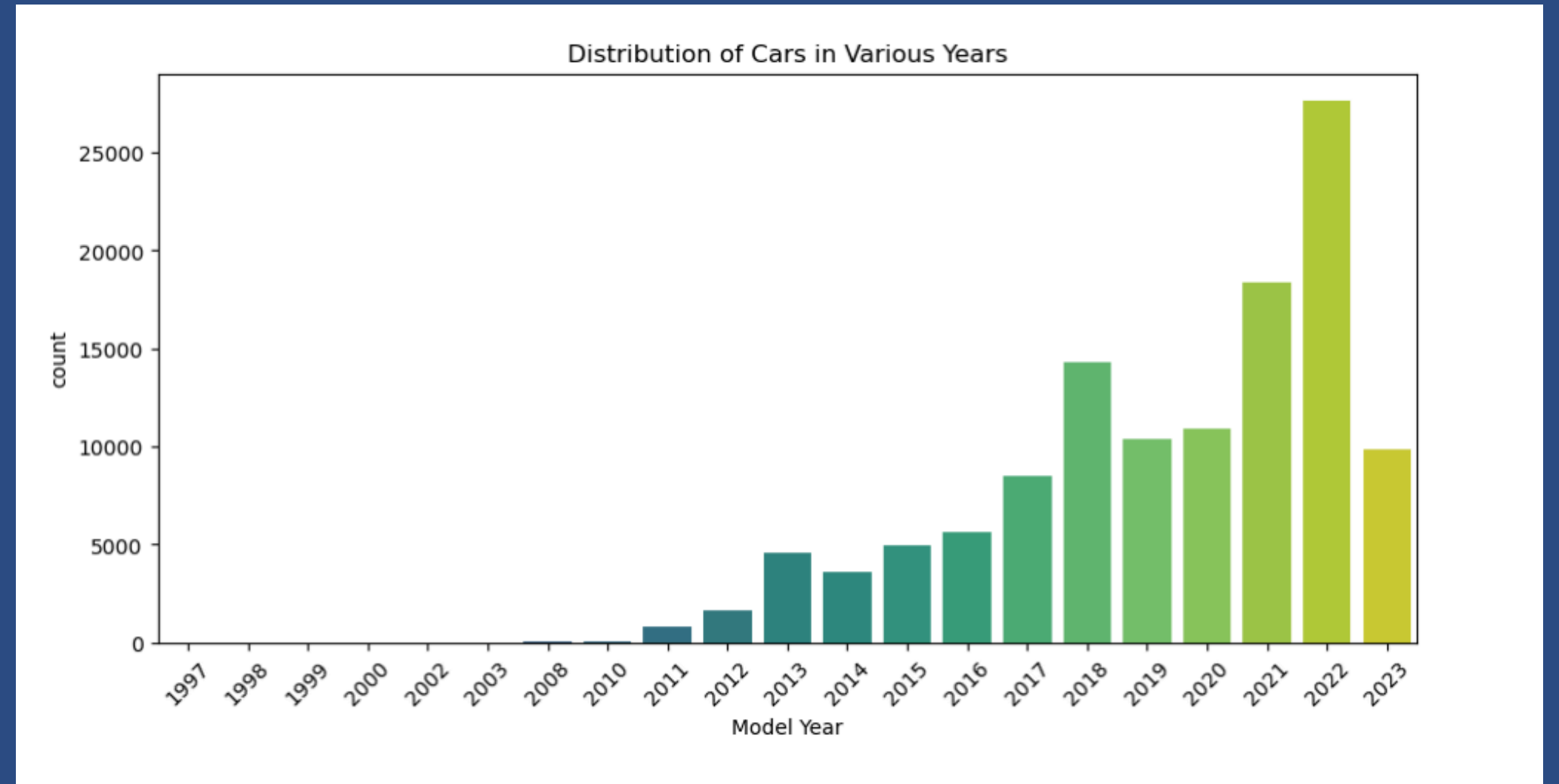
Prior studies used static visuals. Your use of interactive, multi-layered charts (e.g., Kepler.gl, Sankey, Icicle) allows deeper exploration, empowering users to tailor insights to their context.

6

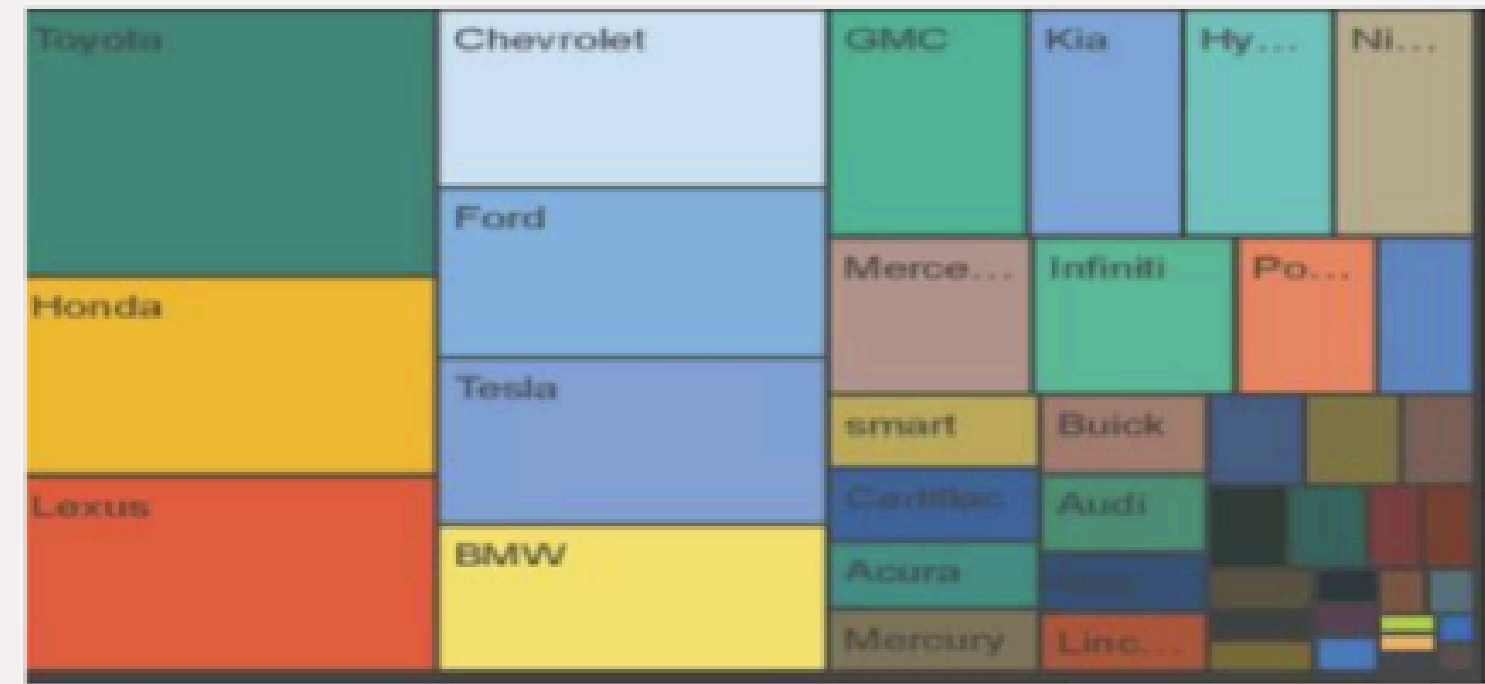
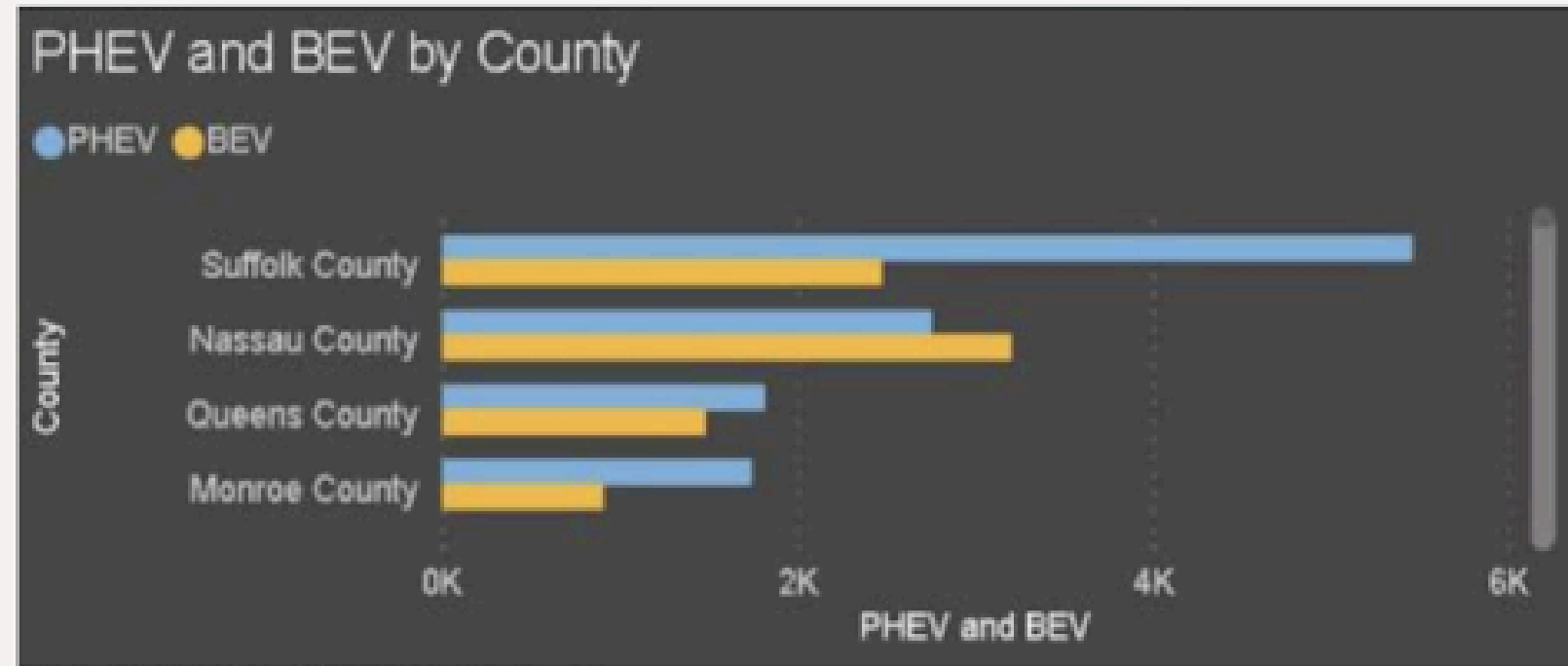
Existing Work



Used basic pie chart and lack of deep insights



The analysis is mostly descriptive and can drill down into each brand and see year wise growth for them.



Grouped bar chart appears cluttered and difficult to navigate.

Tree map visualization for EV models lacks clarity, making it difficult to compare values effectively.

2.2. The county king, Snohomish, Pierce, and Clark have the largest number of electric vehicles

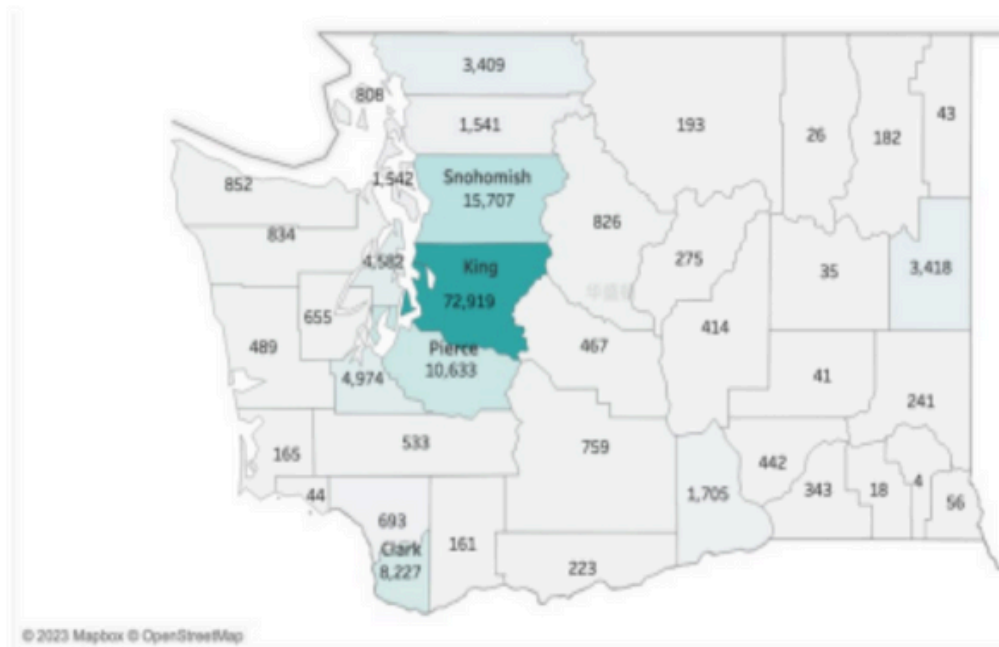
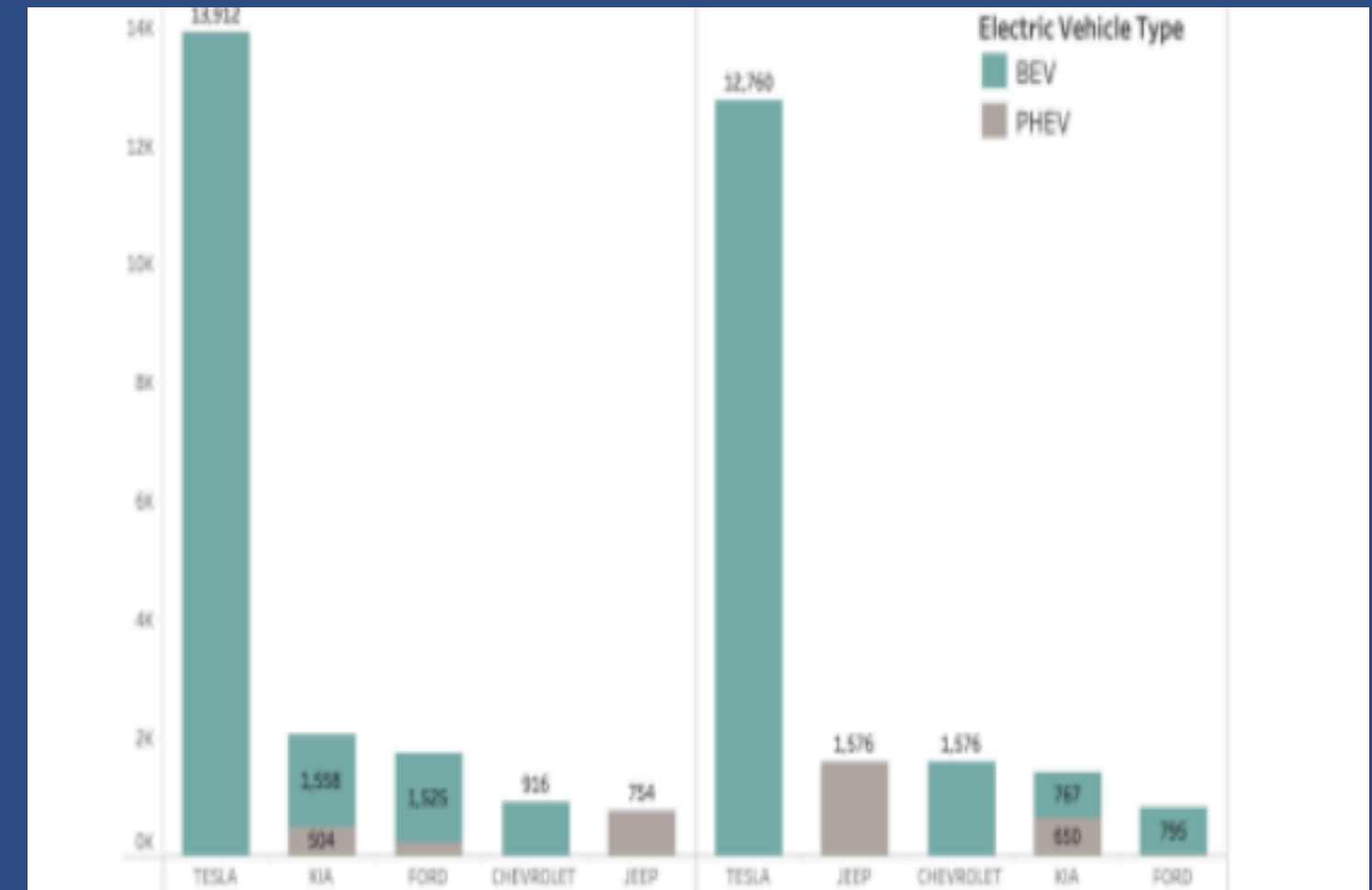


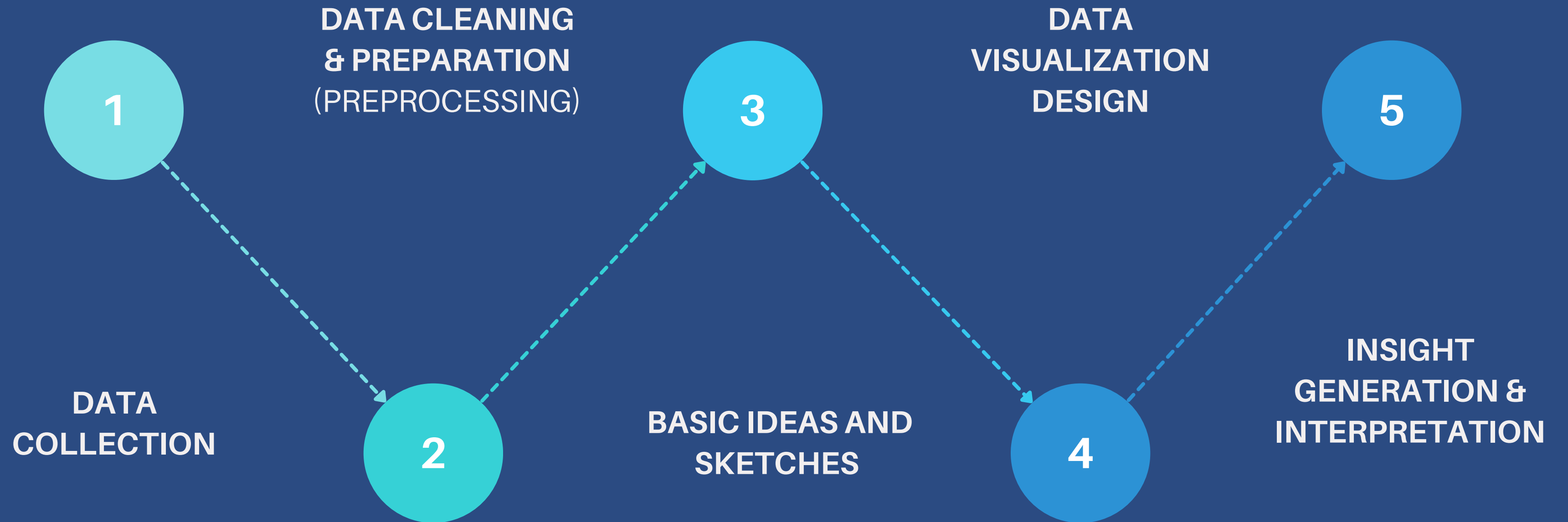
Figure 2: The electric vehicle distribution in Washington by county, including all the electric vehicles manufactured since 1996. The Colour from light to dark shows the sum of electric vehicles in a county.



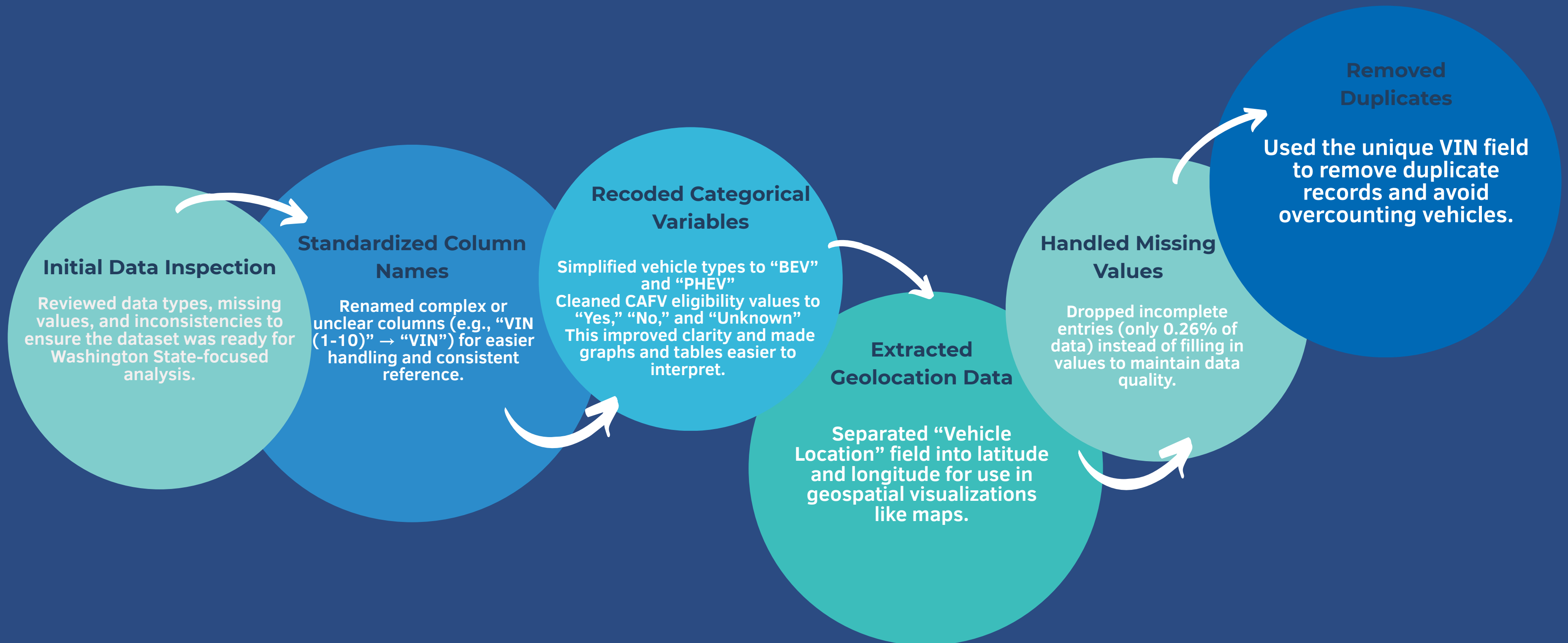
An intensity map would better suit showing the continuous count of EV's manufactured in the top counties.

**Lack of long-term
patterns, doesn't
show fully growth of
EV vehicles**

METHODS



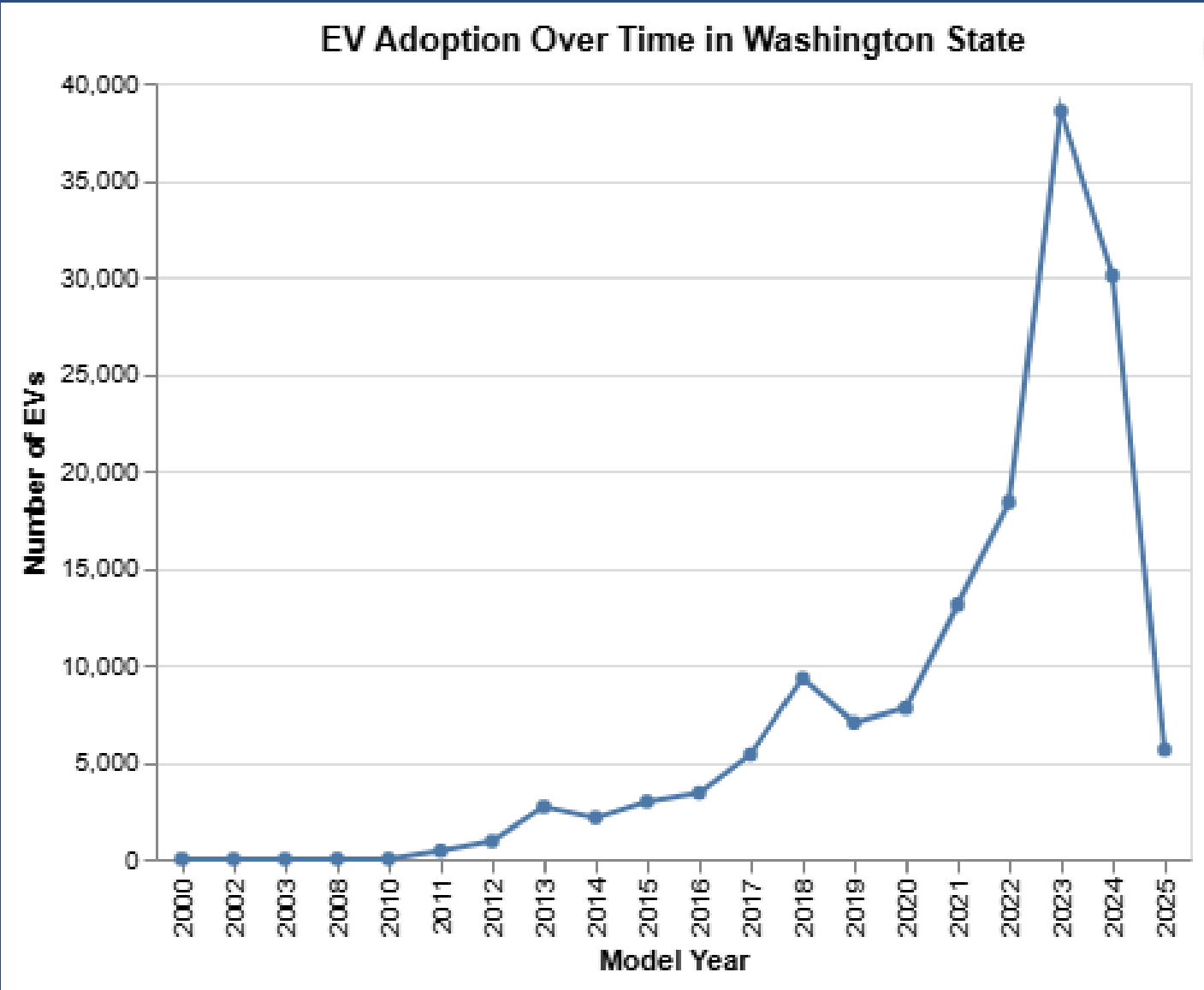
DATA CLEANING AND PREPROCESSING



Initial Sketches and Prototype

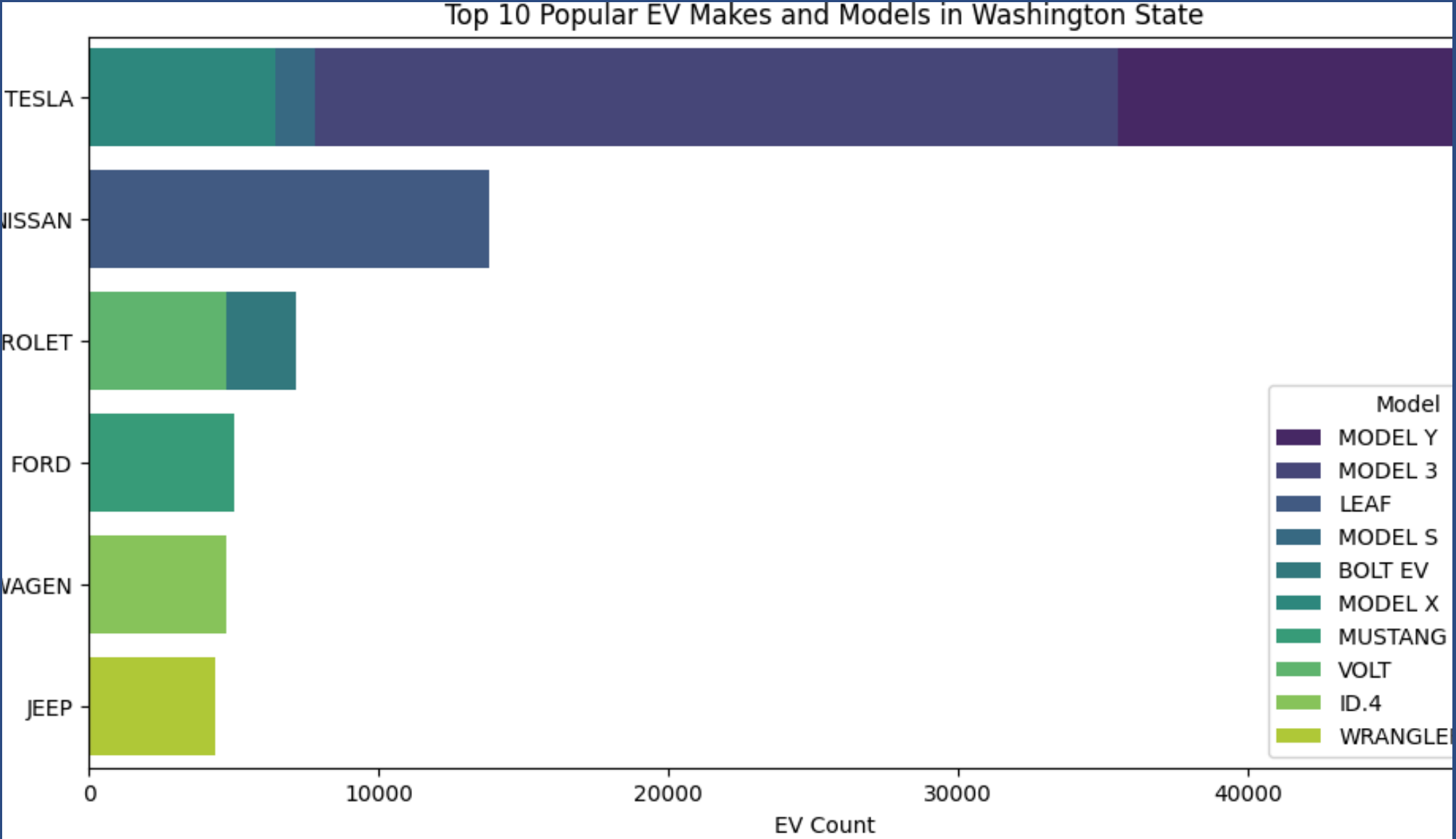
EV ADOPTION OVER TIME

Line charts shows growth trends in EV registrations from 2000–2025, with adoption spikes around major policy and model launches



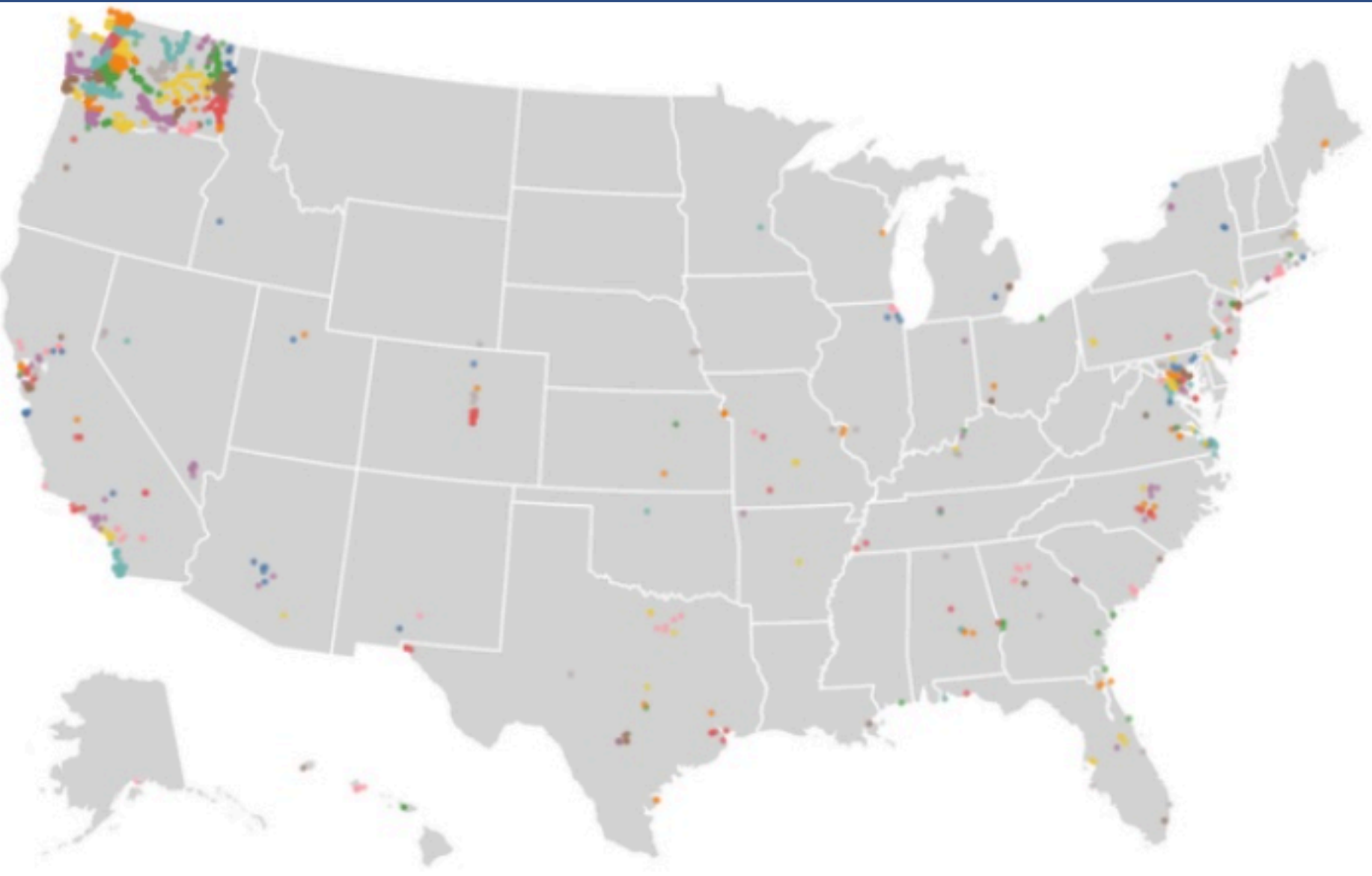
TOP MAKES AND MODELS

Stacked bar charts reveals Tesla, Nissan, and Chevrolet as leaders, with specific models like Model 3 and Leaf dominating within their brands.



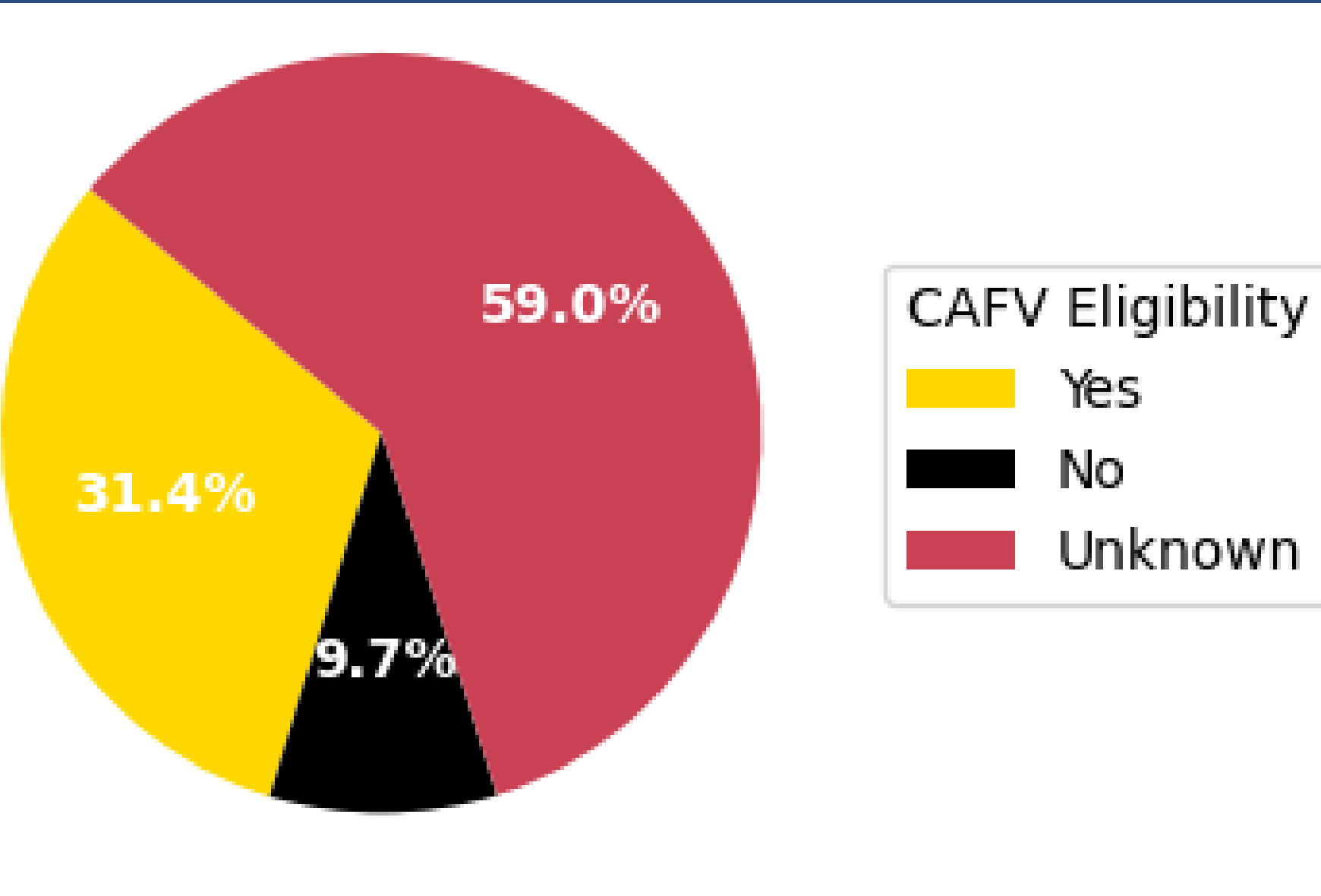
REGIONAL TRENDS

Map and timeline highlighted higher EV concentrations in urban counties like King and Snohomish, with gradual growth in rural areas.



CAFV ELIGIBILITY

Pie chart indicates that most EVs are CAFV-eligible, reflecting strong alignment with clean energy policies.



Key Questions Guiding the Analysis:

1: How has EV adoption changed over time, and how do BEVs and PHEVs compare?

- Identifies trends, growth spikes, and market shifts toward fully electric vehicles in recent years.

3: How do consumer preferences for EV makes and models differ, especially between BEVs and PHEVs?

- Explores brand and model popularity using hierarchical visualizations to show buyer trends.

5: Is there a relationship between MSRP and electric range?

- Tests the assumption that higher price equals better performance, finds minimal correlation.

2: How are EVs distributed across counties in Washington State?

- Reveals regional adoption disparities, highlighting urban centers as adoption hubs.

4: How are MSRP and electric range distributed across EV types?

- Compares pricing and performance, showing BEVs generally cost more but offer longer ranges.

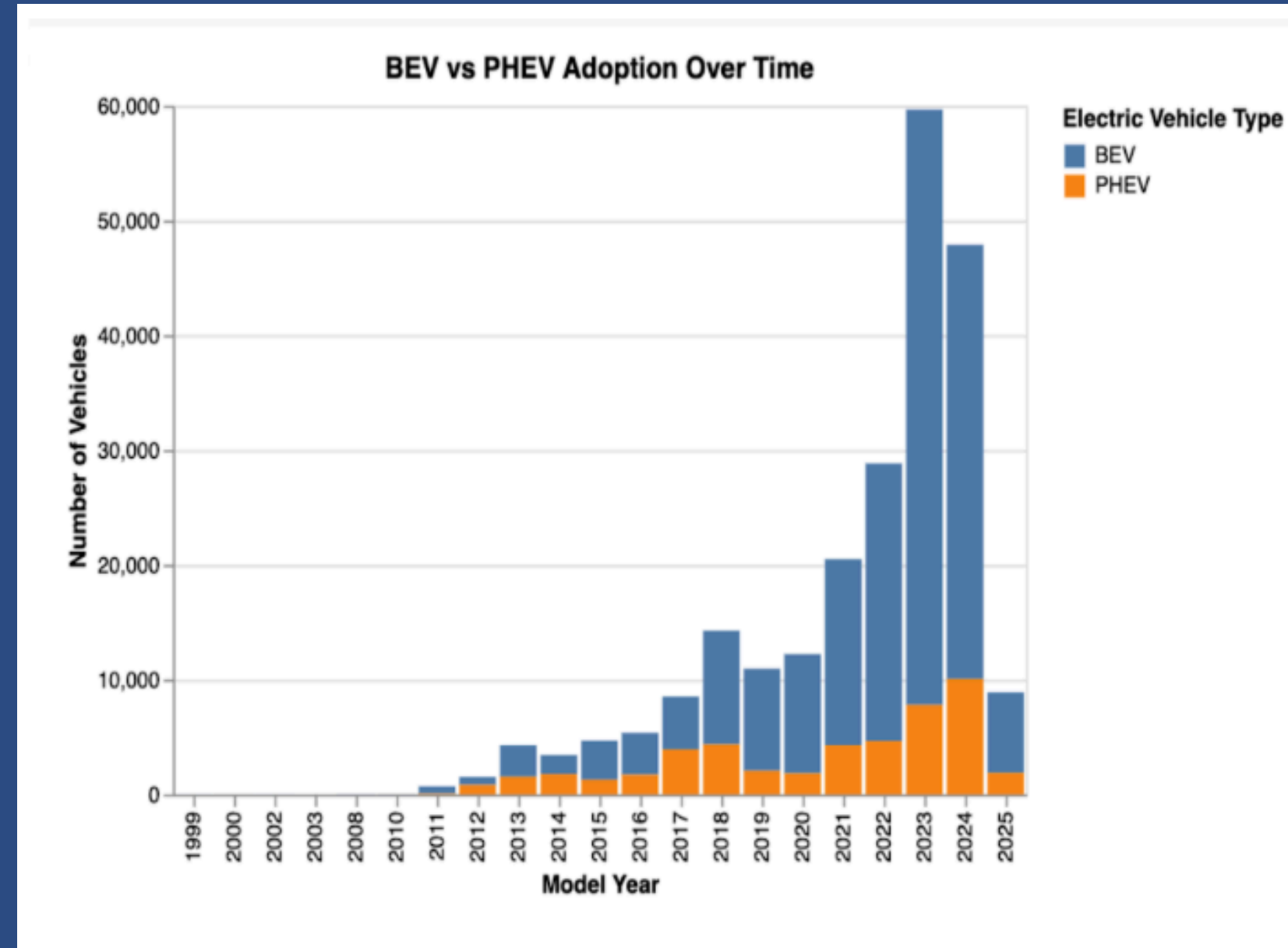
6: How does CAFV eligibility vary across EV models and counties?

- Highlights differences in clean vehicle program coverage by region and model, useful for policy targeting.

Visualizations

EV ADOPTION OVER TIME BY ELECTRIC VEHICLE TYPE

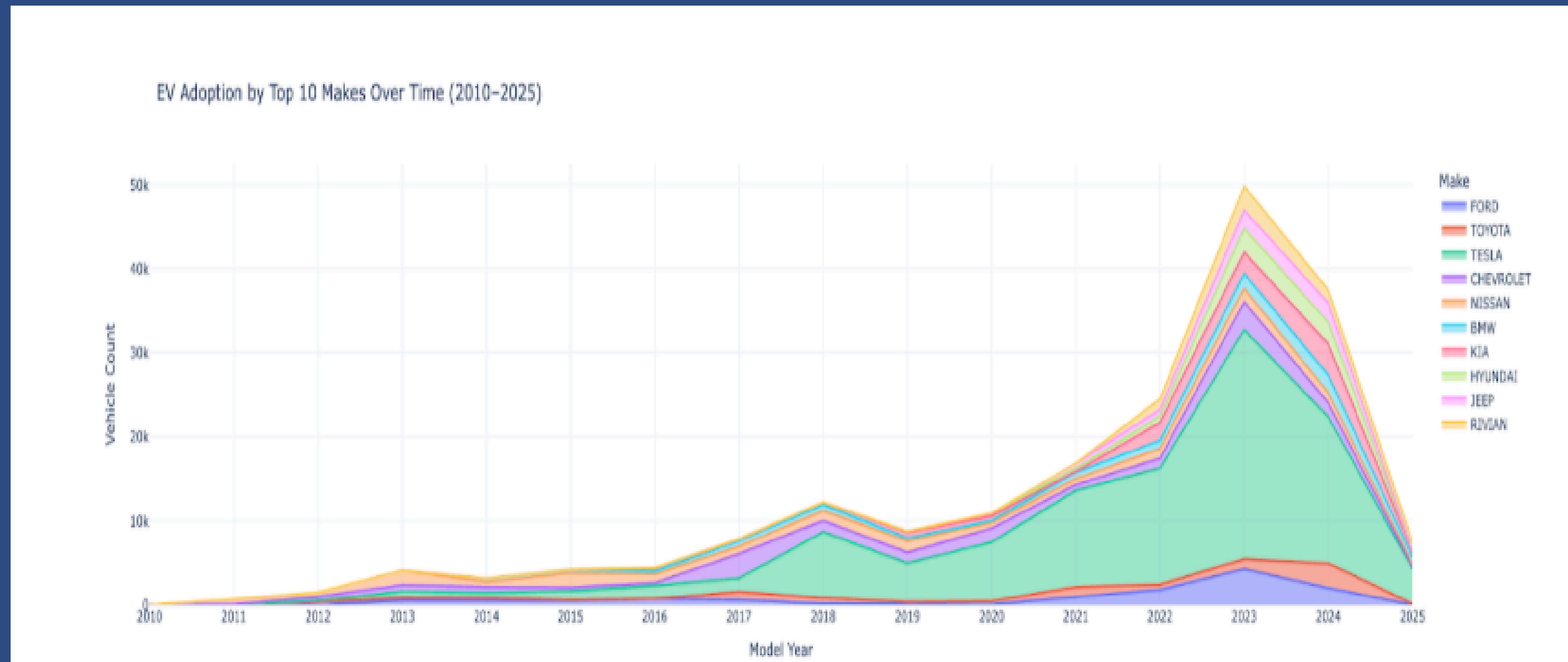
The visualization shows the adoption trends of two types of electric vehicles—Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs)—in Washington State, from 1999 to 2025.



It clearly highlights that consumer preference has shifted significantly toward fully electric vehicles (BEVs), especially since around 2018, with a noticeable peak in 2022–2023. Although PHEVs have been adopted, their growth remains considerably smaller in comparison.

TIME SERIES ANALYSIS OF TOP 10 MAKES

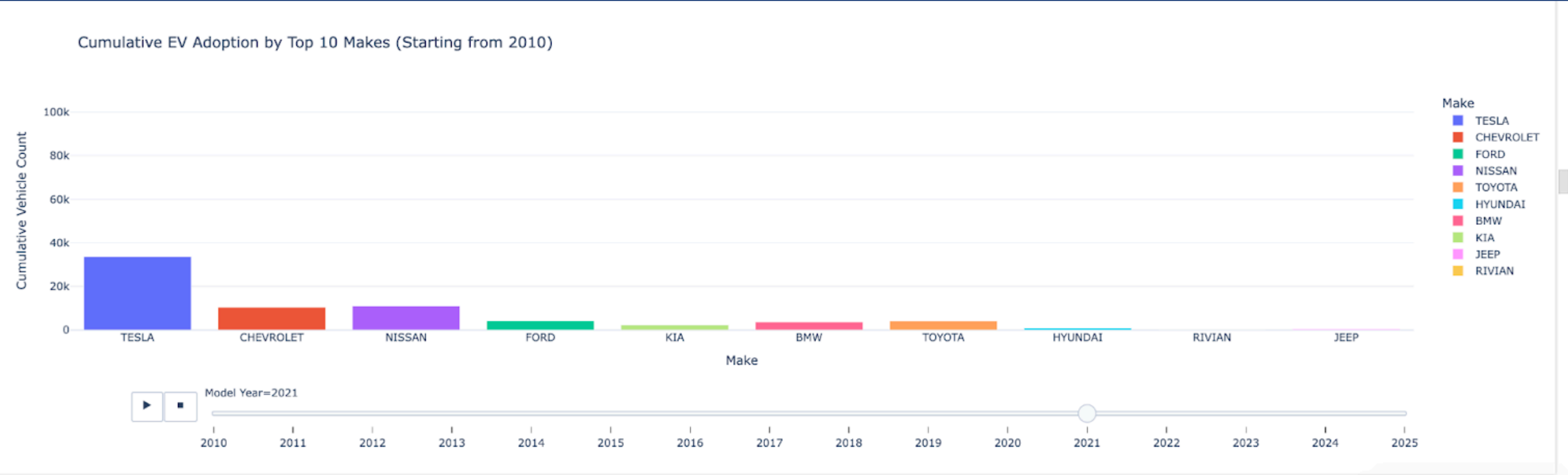
The visualization shows the adoption trends of electric vehicles (EVs) by the top 10 most popular automakers in Washington State, from 2010 to 2025.



It reveals significant growth in EV adoption across various automakers, particularly highlighting Tesla and Nissan as dominant brands in recent years. The chart effectively shows both individual brand growth and the cumulative increase in EV popularity, offering insights into market competition and shifts in consumer brand preferences.

CUMULATIVE EV ADOPTION BY TOP 10 MAKES (2010–2025)

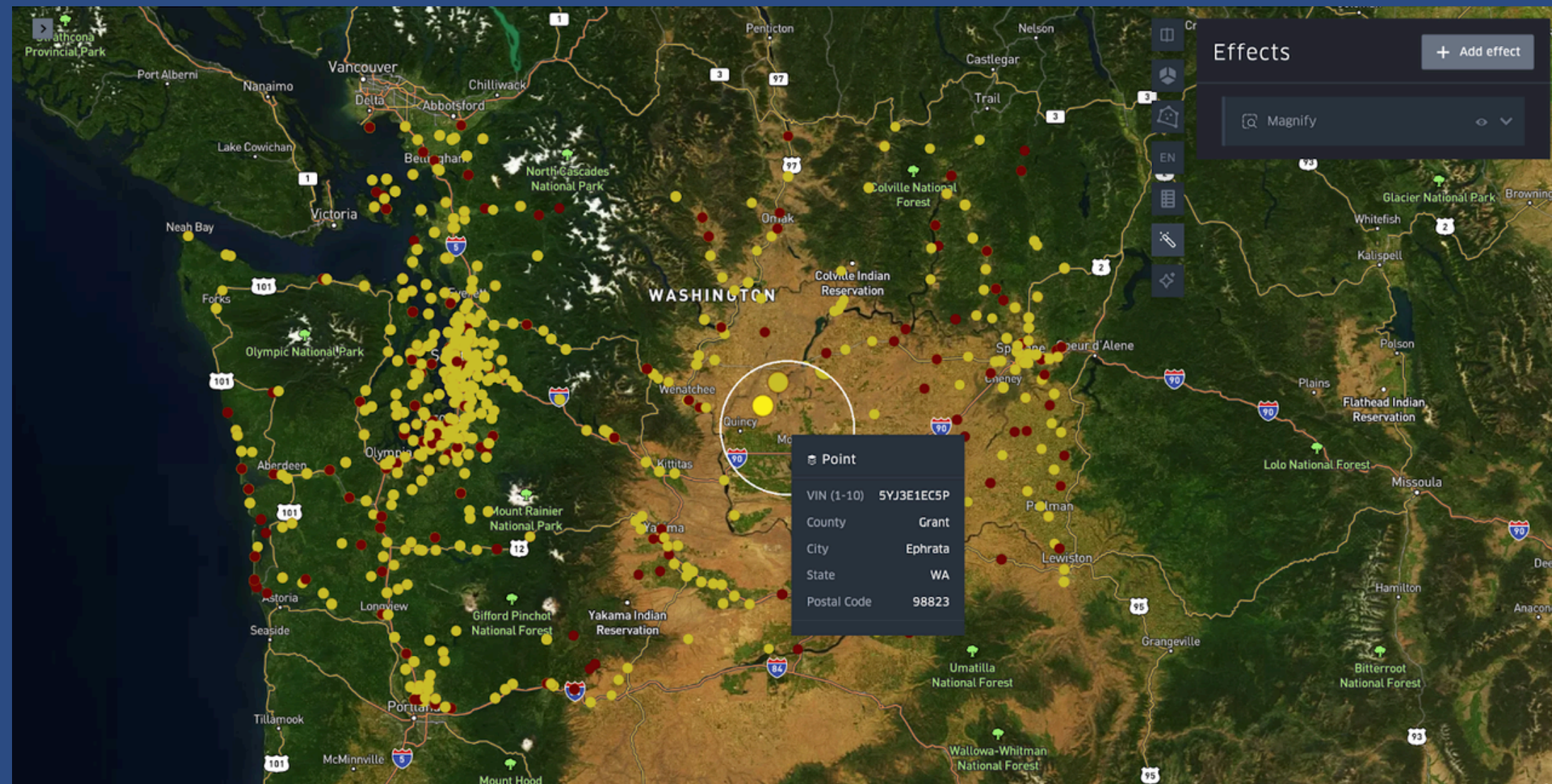
Visualize how EV adoption has progressed over time for the 10 most popular makers in Washington State using an animated bar chart.



Tesla and Nissan lead in cumulative EV adoption. Reveals brand-wise growth patterns and market shifts over time. Useful for understanding consumer behavior, manufacturer trends, and policy impact on EV adoption.

INTERACTIVE EV DISTRIBUTION VISUALIZATION USING KEPLER.GL

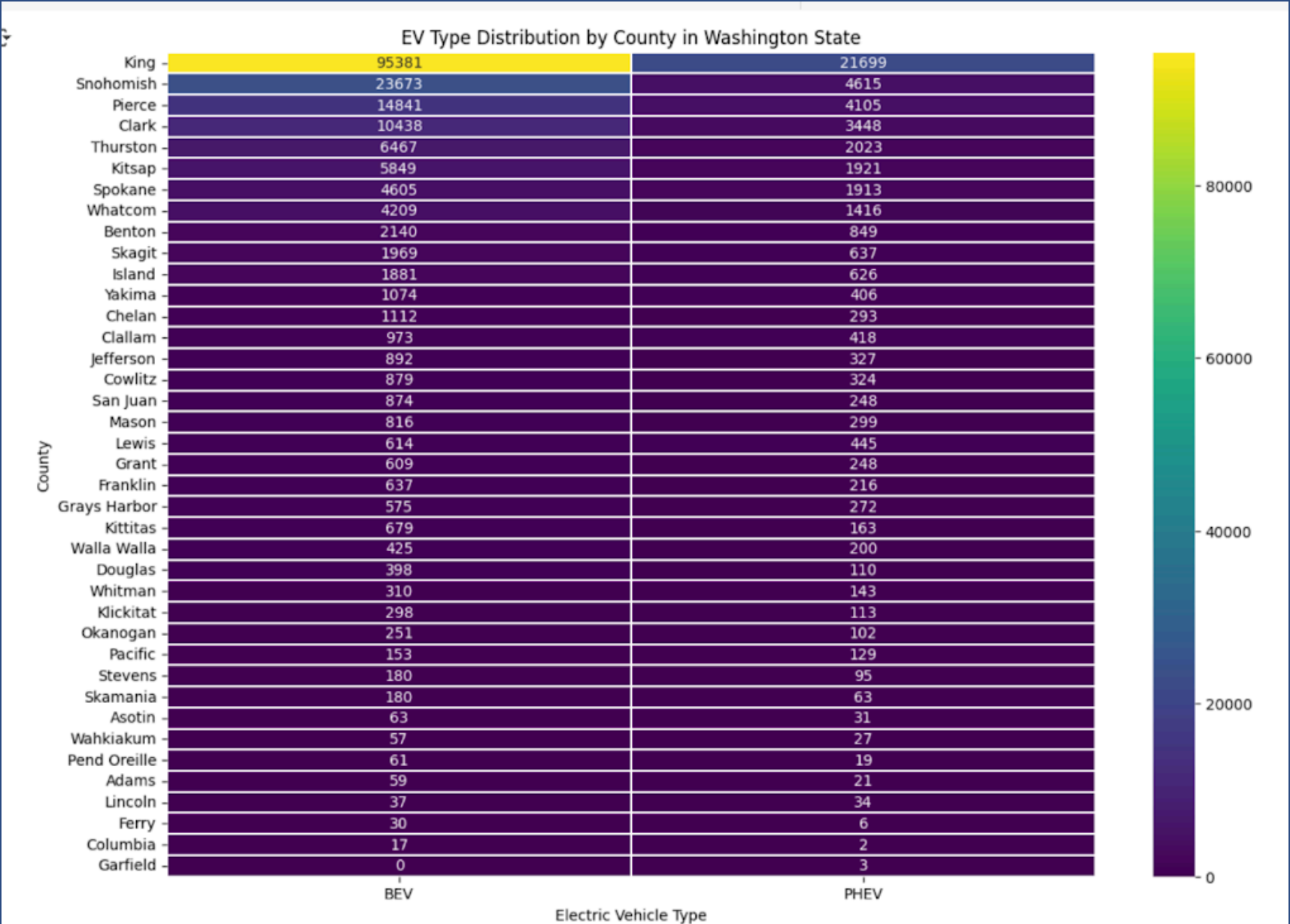
This interactive map shows the distribution of electric vehicles (BEVs and PHEVs) across Washington State using color-coded markers.



It reveals that EV adoption is much higher in urban areas—especially around Seattle—compared to rural regions, showing clear spatial patterns of where EVs are most common

EV DISTRIBUTION BY COUNTY

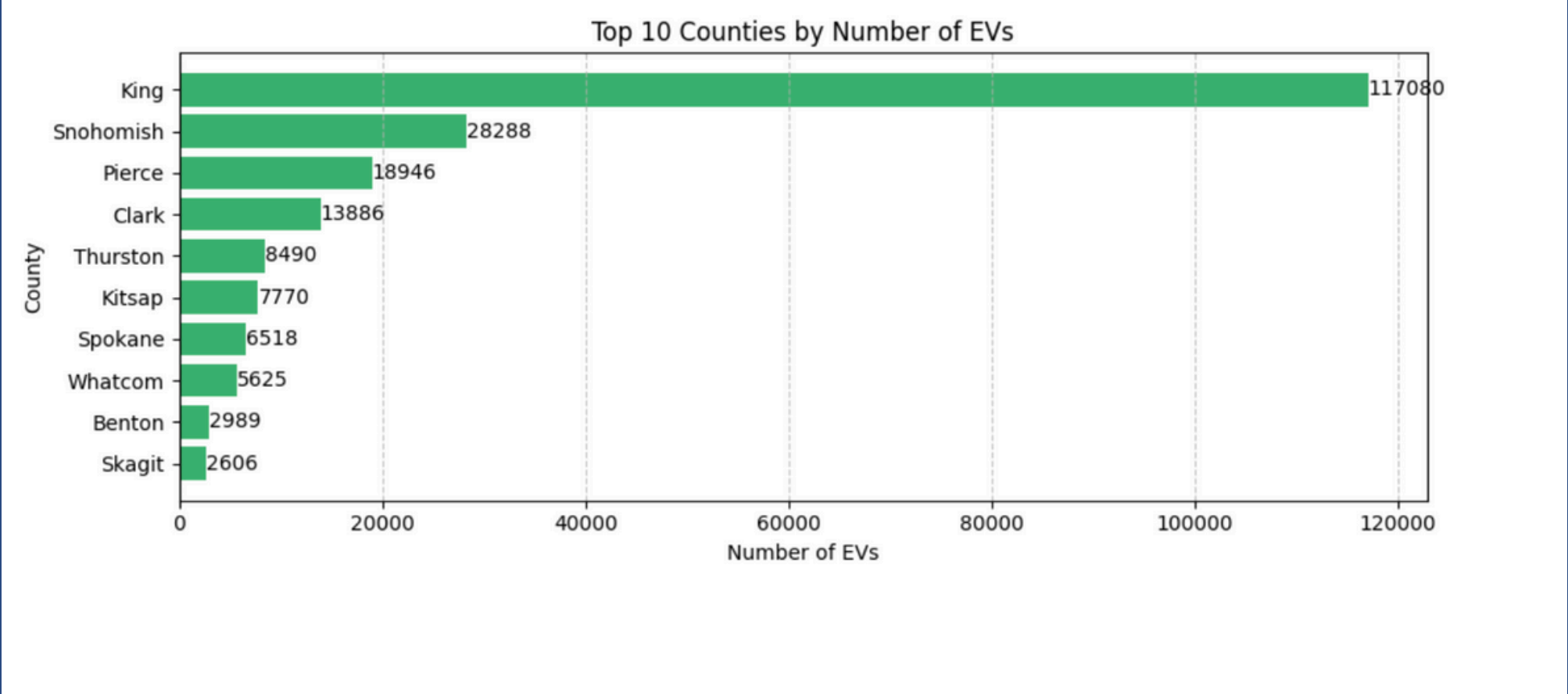
This heatmap shows the distribution of BEVs and PHEVs by county in Washington State, using color intensity to represent vehicle counts.



It highlights that EV adoption is highest in urban counties like King and lowest in rural areas, offering a clear visual comparison across the state.

TOP 10 COUNTIES AND NUMBER OF EV'S

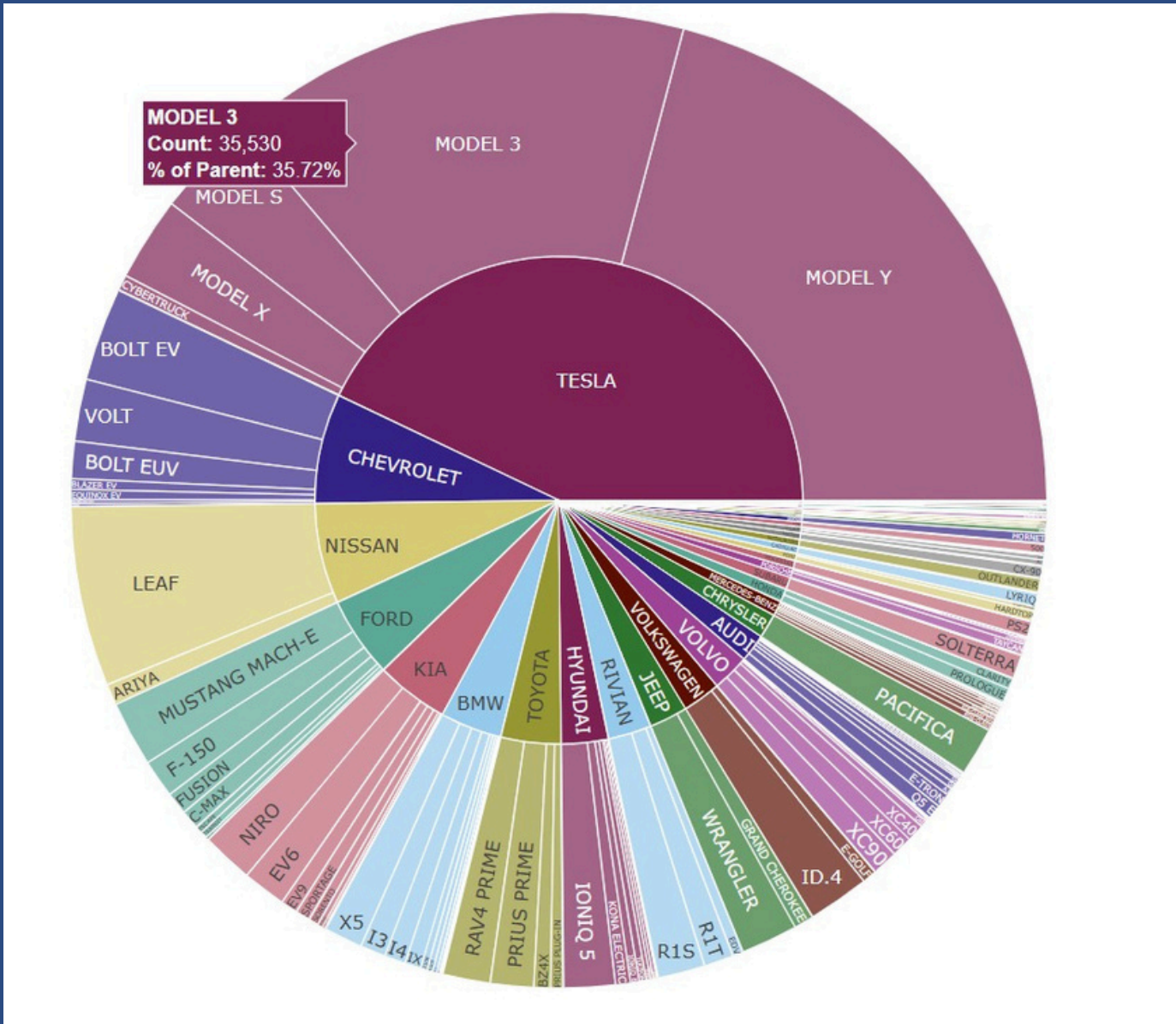
This horizontal bar chart displays the number of electric vehicles across the top counties in Washington State.



It shows that King County leads by a large margin in EV adoption, with a sharp drop-off in other counties, clearly highlighting regional disparities.

EV DISTRIBUTION BY MAKE AND MODEL USING SUNBURST CHART

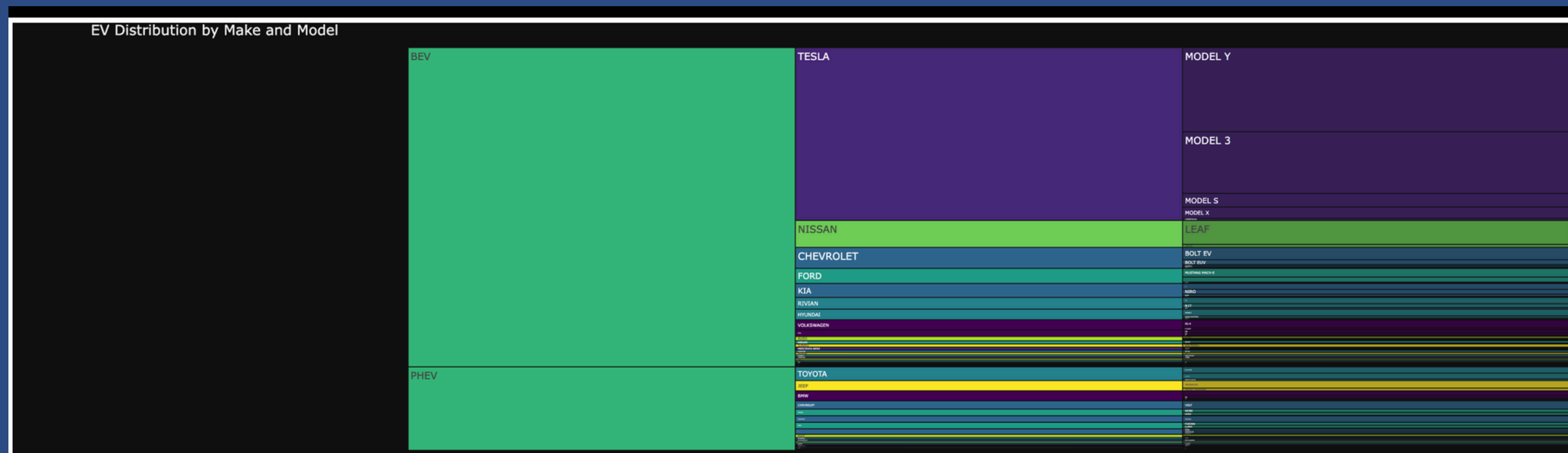
This Sunburst chart shows the hierarchy of electric vehicle adoption, starting from vehicle make and drilling down to specific models.



It reveals which brands dominate the EV market and how diverse their model offerings are, helping to highlight both market leaders and model variety in a visually intuitive way.

EV DISTRIBUTION BY MAKE AND MODEL

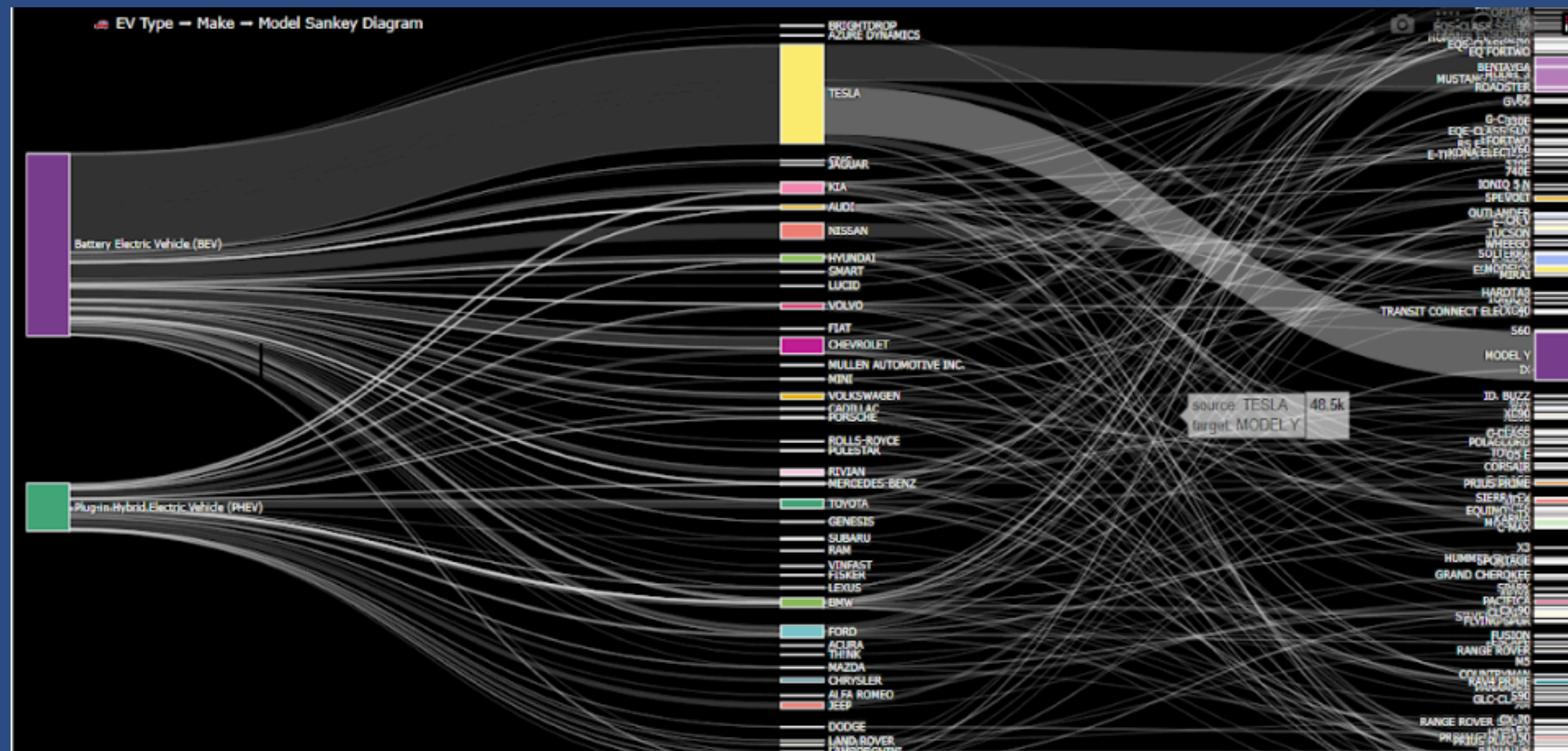
This icicle chart visualizes the hierarchical distribution of electric vehicles—from EV type (BEV or PHEV) to manufacturer (make) and model—allowing users to explore the structure of the EV market.



It helps identify which makes dominate specific EV types, how diversified each make's offerings are, and which models are most popular, giving a clear view of market structure and brand positioning.

ELECTRIC VEHICLE FLOW: TYPE TO MAKE TO MODEL (TOP 10 MAKES)

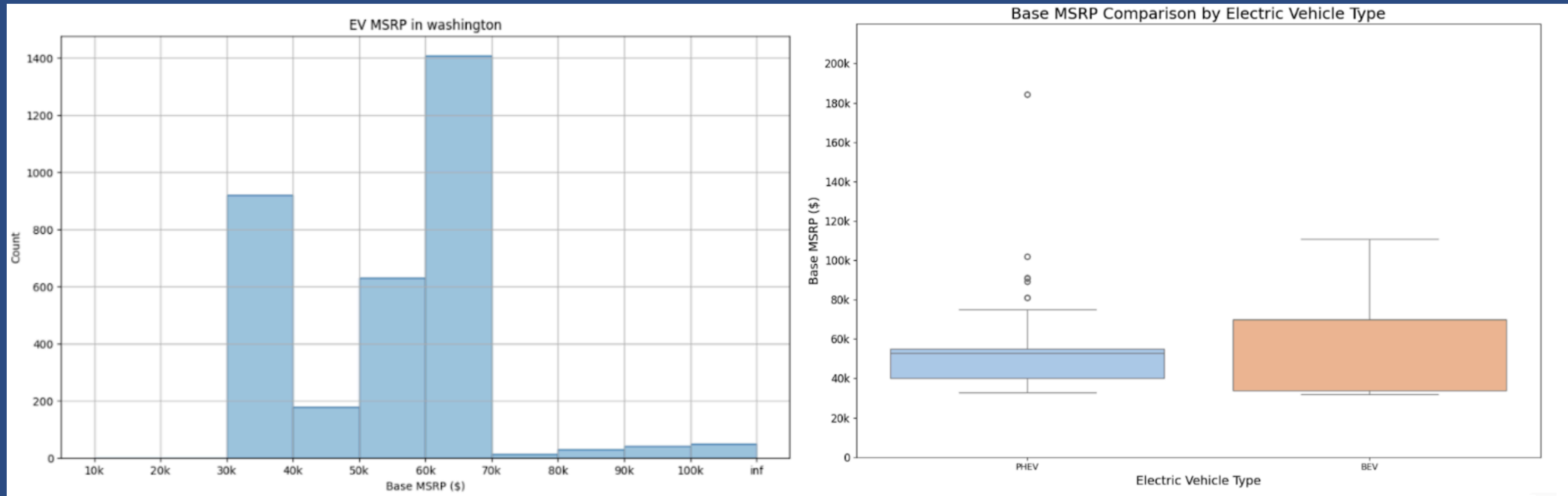
This Sankey diagram visualizes the flow of electric vehicles from type (BEV or PHEV) to make and then model, showing how the EV market is structured across these categories.



It highlights which EV types are most popular, which makes dominate each type, and which models are most common—providing a clear picture of market concentration and diversity across the EV landscape.

BASE MSRP ACROSS EV BY EV TYPE

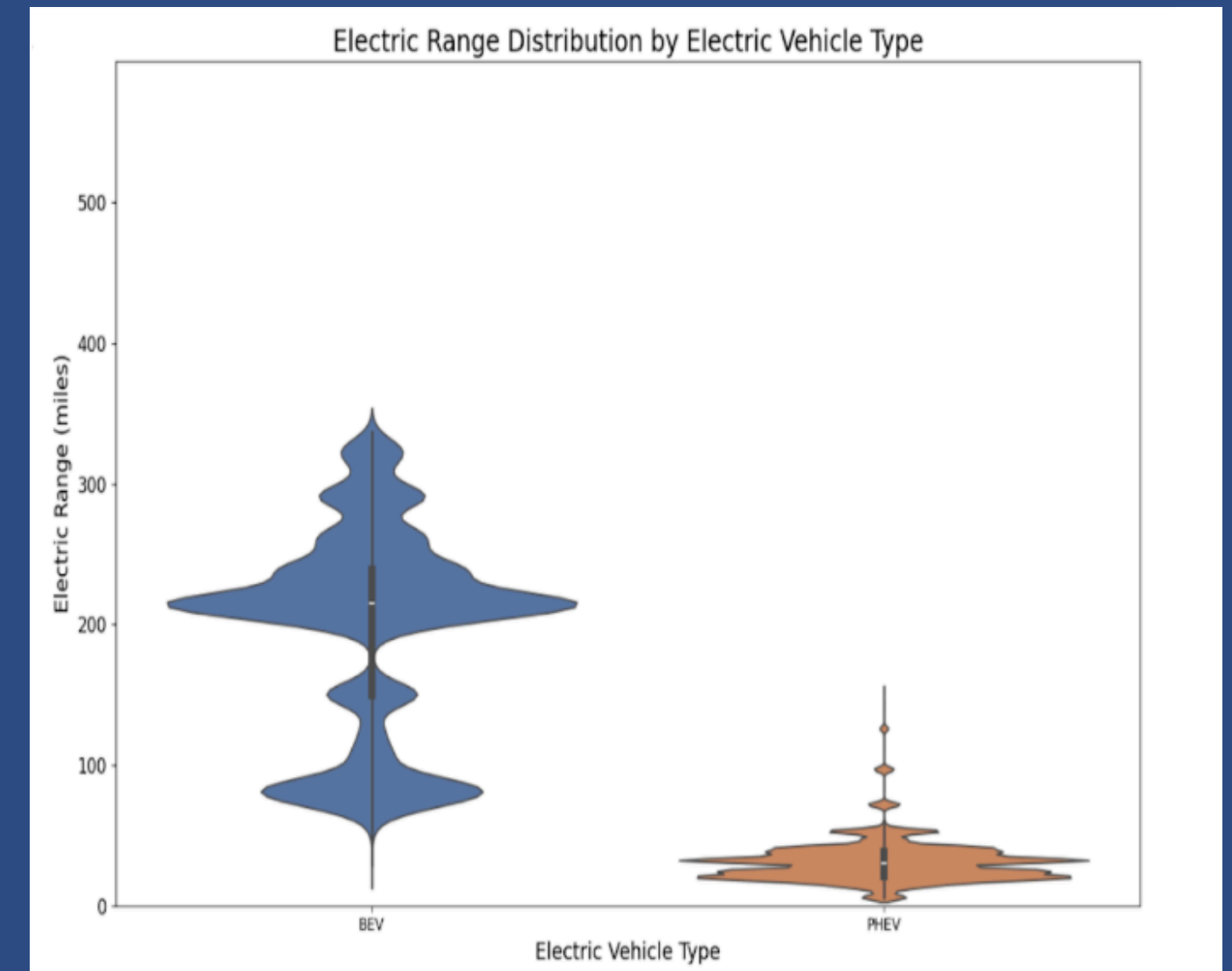
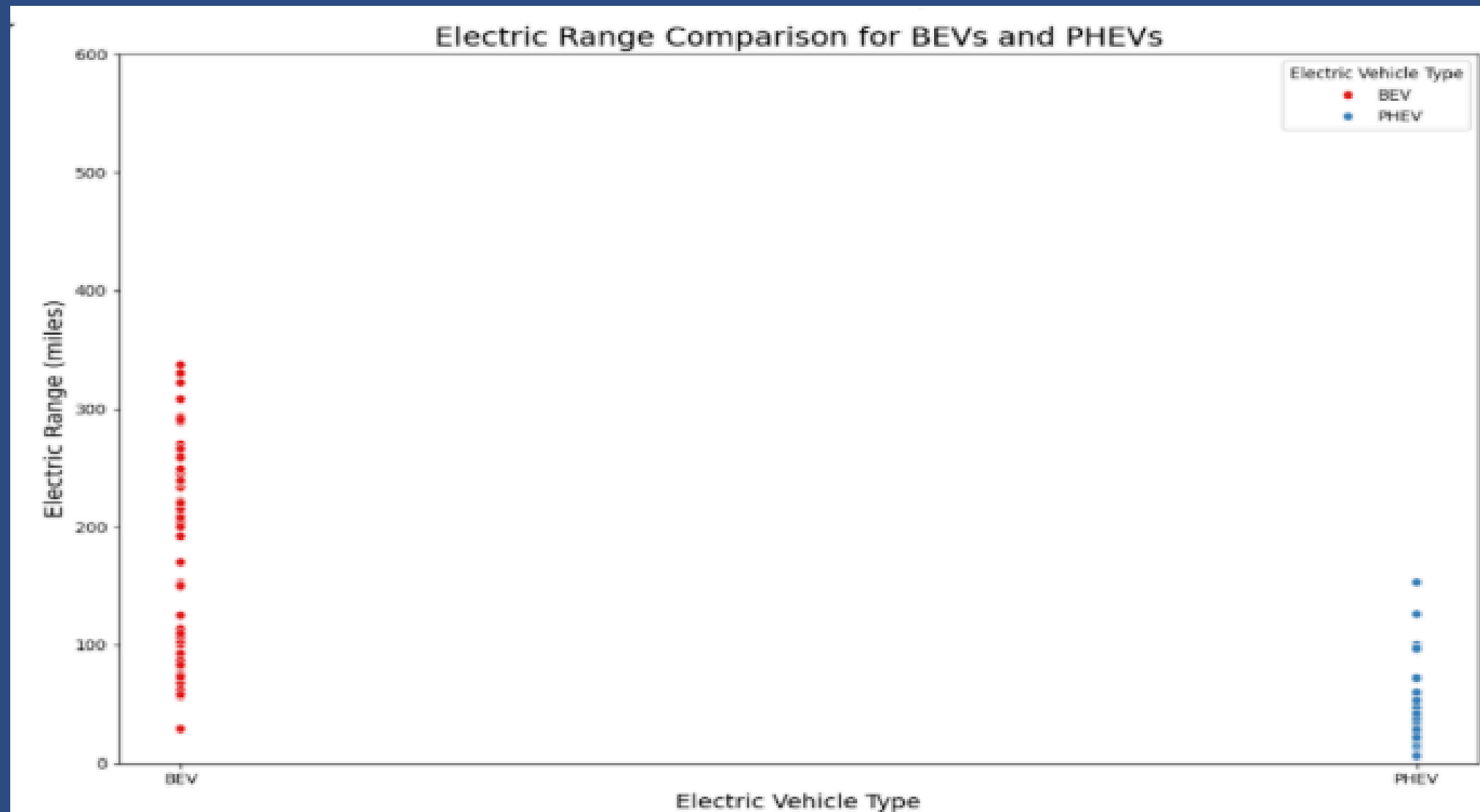
The histogram displays how electric vehicles are distributed across different MSRP price ranges in Washington State. This box plot compares the base MSRP of Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs).



It shows that most EVs fall within the \$30,000–\$70,000 range, especially around \$60,000–\$70,000, suggesting a market preference for mid-range over luxury or budget models and BEVs tend to be more expensive and have a wider price range than PHEVs, with more high-end outliers, highlighting key price differences between the two EV types.

ELECTRIC RANGE DEPENDING ON THE EV TYPE

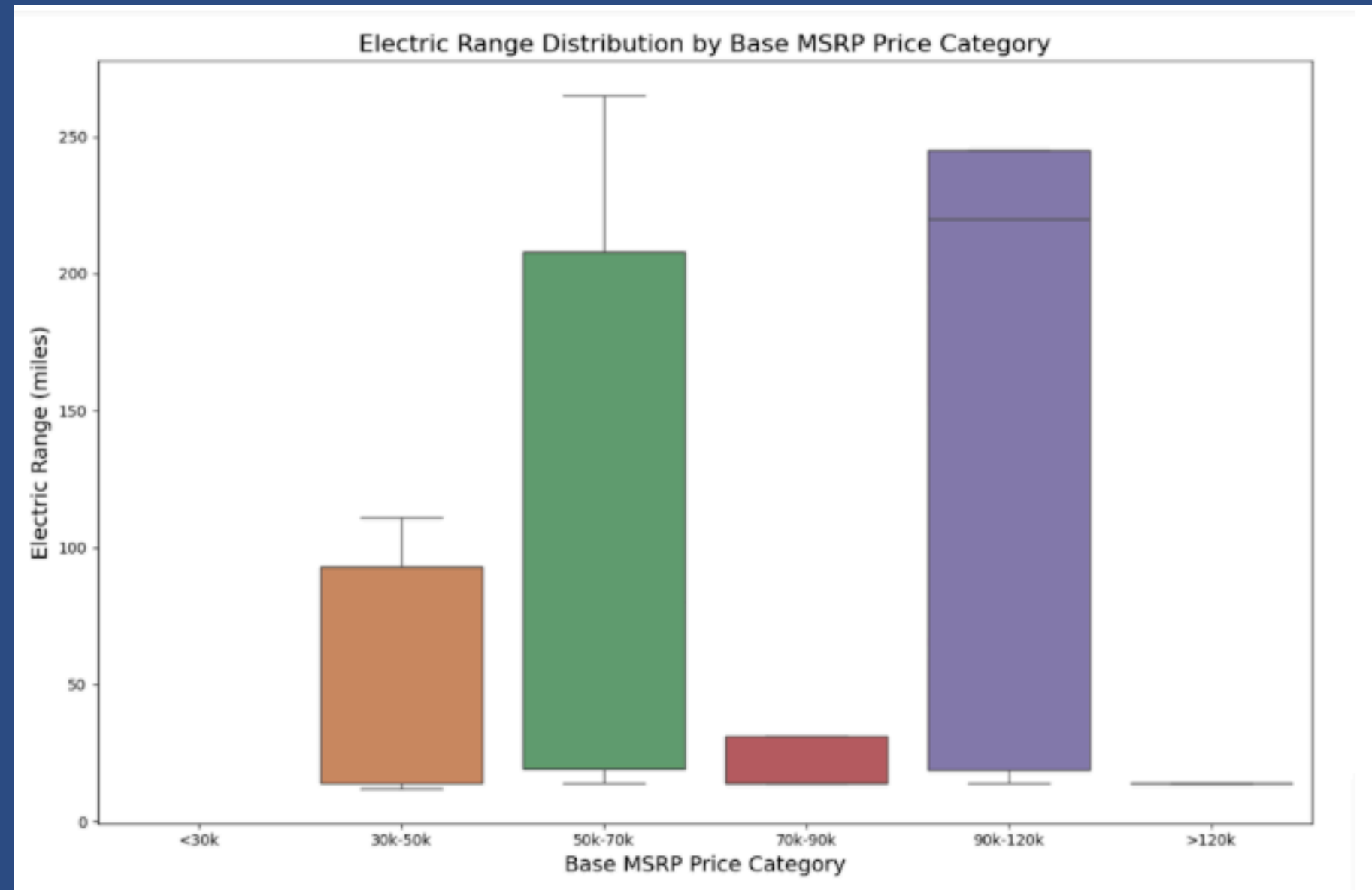
This scatter plot compares the electric driving ranges of Battery Electric Vehicles (BEVs) and Plug-in Hybrid Electric Vehicles (PHEVs).



It shows that BEVs have significantly higher and more varied electric ranges than PHEVs, which mostly cluster below 50 miles—clearly illustrating the performance gap in electric-only driving capability.

ELECTRIC RANGE DEPENDING ON THE VEHICLE PRICE

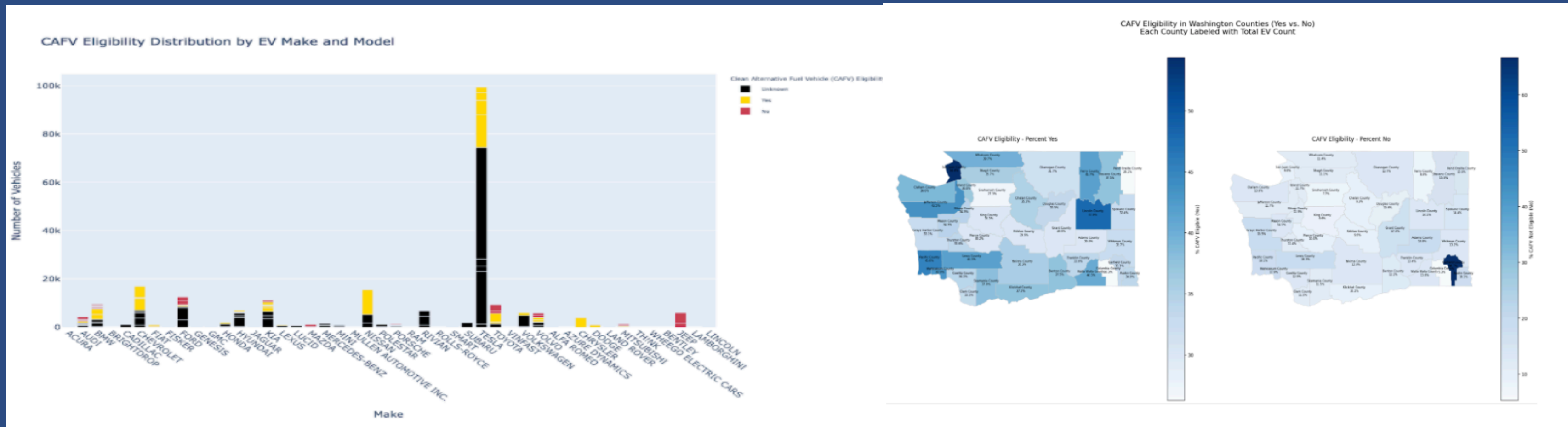
THIS BOX PLOT COMPARES THE ELECTRIC DRIVING RANGE OF EVS ACROSS DIFFERENT MSRP (PRICE) CATEGORIES.



It shows that vehicles priced between \$50k–\$100k offer the highest average electric ranges, while some premium models above \$120k offer lower ranges indicating that a higher price does not always guarantee better driving performance.

CAFV ELIGIBILITY BY MODEL AND COUNTY

These charts explore how Clean Alternative Fuel Vehicle (CAFV) eligibility varies across EV makes, models, and counties in Washington State.



- The bar chart shows that major brands like Tesla and Nissan dominate the CAFV-eligible vehicle counts, with most of their models marked as "Yes" for eligibility. However, several other brands have mixed or unknown statuses.
- The choropleth maps reveal geographical disparities urban counties like King and Snohomish lead in both total eligible EVs and percentage-wise coverage, while many rural counties lag behind.

KEY TAKEAWAYS

Geographic Insights:

Spatial Exploration: Analyzes EV adoption trends at the county level, identifying regions with higher adoption and infrastructure gaps.

Interactive Exploration:

Kepler.gl Maps: Enables spatial exploration with pan, zoom, and filter options at the county level for deep geographic analysis
Plotly Visualizations: Interactive charts with hover tooltips, filters, and animated timelines to engage both technical and non-technical users dynamically..

Temporal Patterns

Animated EV Adoption Over Time by Make (Area Chart): Shows growth patterns among top manufacturers (e.g., Tesla, Chevrolet) with year-over-year momentum. EV Count by Model Year (Line Chart): Highlights key adoption periods, influenced by policy incentives or infrastructure expansions.

Policy & Technical Characteristics:

CAFV Eligibility by Make and Model (Stacked Bar Chart): Reveals eligibility variations by brand and model, highlighting trends and potential policy gaps.

CAFV Eligibility (Pie Chart): Simplified, high-level distribution of eligibility statuses for quick audience understanding.

LIMITATIONS

- Missing and ambiguous data (e.g., CAFV eligibility marked "unknown") reduces interpretability and weakens correlation insights.
- Lack of population normalization may skew county-level analysis, overrepresenting high-population regions.

FUTURE RECOMMENDATIONS

- **Population-Normalized Metrics:** Incorporating county-level population or vehicle registration data to reveal true regional disparities in EV adoption.
- **Demographic Integration:** Linking EV data with demographic factors (income, education, urban-rural classification) to enable more equitable policy interventions.
- **Infrastructure Readiness:** Assessing the availability of charging stations, electric utility coverage, and grid capacity to ensure regions are ready for increased EV demand.
- **Temporal Forecasting:** Using historical trends in registration, eligibility rates, and manufacturer growth to predict future EV adoption patterns under various policy scenarios.
- **Goal:** Improve the quality, equity, and forecasting of future EV adoption trends to inform more effective planning and policy decisions.

References

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Video Link and Project Work



https://youtu.be/D7_-ZHpnmY8



<https://github.com/prachi1211/Data-Visualization-Project>